

Due: Thursday, October 6th at the beginning of class. But it is strongly suggested you complete this before exam 1 on September 28th.

Read Section 1.1 of *Levin* before you start this homework. Then complete the following:

- Chapter 1, Section 1: # 1 - 6, 9, 10, (These do not require explanation, except for # 10.)
- And complete the following:

Consider a standard deck of 52 playing cards as our universe. Explain your answer fully.

- (1) Let C be the set of cards that are in the suite clubs. What is $|C|$?
- (2) Let F be the set of face cards. What is $|F|$?
- (3) How many cards are clubs and face cards?
- (4) How many cards are clubs or face cards?
- (5) How many ways can I choose one card from the deck?
- (6) How many ways can I choose one card, replace it, shuffle, and then choose a second card?
- (7) How many ways can I choose two cards from the deck? (A two card hand, no replacement.)
- (8) How many ways can I choose two clubs from the deck, without replacement?
- (9) How many ways can I choose two cards from the deck, where the first is a club?
- (10) How many ways can I choose two cards from the deck, where exactly one is a club?
- (11) How many ways can I choose two cards from the deck, where both are clubs?
- (12) How many ways can I choose two cards from the deck, where one is a club and one is a face card?